# Teaching and Examination Regulations

## Faculty of Science
## Master's programme in Computer Security

### Academic year 2025-2026

A. Faculty section
B1. Programme-specific section – general provisions
B2. Programme-specific section – content of programme

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## Section A: Faculty section

### 1. General provisions

#### Article 1.1 Applicability of the Regulations

| | | |
|---|---|---|
| 1. | These Regulations apply to anyone enrolled in the programme, irrespective of the academic year in which the student was first enrolled in the programme. | Advice OLC, approval FGV (9.38 sub b) |
| 2. | These Regulations enter into force on 1 September 2025. | Advice OLC, approval FGV (9.38 sub b) |
| 3. | An amendment to the Teaching and Examination Regulations is only permitted to concern an academic year already in progress if this does not demonstrably damage the interests of students. | Advice OLC, approval FGV (9.38 sub b) |

#### Article 1.2 Definitions

The following definitions are used in these Regulations (*in alphabetical order*):

a. academic year: the period beginning on 1 September and ending on 31 August of the following calendar year;

b. EC (European Credit): a course credit with a workload of 28 hours of study;

c. examination: an assessment of the student's knowledge, understanding and skills relating to a unit of education. The assessment is expressed in terms of a final mark. An examination may consist of one or more partial examinations. A resit always covers the same material as the original examination. An examination can be completed in writing, orally or in another way;

d. final examination a Master's programme consists of a Master's examination;

e. internship period spent working in professional practice as part of a study programme;

f. 1. joint degree: a degree awarded by an institution together with one or more institutions in the Netherlands or abroad, after the student has completed a study programme (a degree programme, a major or a specific curriculum within a degree programme) for which the collaborating institutions are jointly responsible;

2. double degree: partnership between a programme of a Dutch educational institution with a foreign educational institution, whereby students complete all or part of both study programmes and obtain two diplomas on the basis of agreements regarding the mutual recognition of the programme (replacement courses);

g. period: a part of a semester;

h. practical exercise: participation in a practical training or other educational learning activity, aimed at acquiring certain (academic) skills. Examples of practical exercises are:

- researching and writing a thesis or dissertation
- carrying out a research assignment
- taking part in fieldwork or an excursion
- taking part in another educational learning activity aimed at acquiring specific skills, or
- participating in and completing a work placement;

i. programme: the totality and cohesion of the units of education, teaching activities/methods, contact hours, testing and examination methods and recommended literature;

j. SAP/SLM: the student information system (*Student Lifecycle Management*);

| | | |
|---|---|---|
| k. | semester: | the first (September - January) or second (February - August) half of an academic year; |
| l. | specialisation: | optional route of study within a degree programme indicating a deepening of the context of the programme (e.g. interdisciplinary or multidisciplinary); |
| m. | student | person studying; |
| n. | student statute: | sets out the rights and responsibilities of students on the one hand, and of Vrije Universiteit Amsterdam on the other hand, including those derived from the law and those derived from university regulations. The Executive Board (CvB) officially confirms the student statute once its completeness has been approved by the University Student Council (USC); |
| o. | study guide: | the guide for the study programme that provides further details of the courses, provisions and other information specific to that programme. The Study Guide is available electronically at: https://studiegids.vu.nl/en#/; |
| p. | study monitor: (*studiemonitor*): | dashboard for students and academic advisors containing data pertaining to the student, including the student's study progress; |
| q. | subject: | see t. 'unit of education'; |
| r. | thesis | a unit comprising research into the literature and/or contributing to scientific research, always resulting in a written report; |
| s. | track: | a study pathway within a broader Bachelor's or Master's degree programme, such as a fully English-language study pathway within a Dutch-language Bachelor's or Master's programme; |
| t. | unit of education: | a unit of study of the programme within the meaning of the WHW; |
| u. | university: | Vrije Universiteit Amsterdam; |
| v. | WHW: | the Dutch Higher Education and Research Act (Wet op het Hoger Onderwijs en Wetenschappelijk Onderzoek); |
| w. | workload: | the workload of the unit of education to which an examination applies, expressed in terms of credits = EC credits (ECTS = European Credit and Transfer Accumulation System). The workload for 1 year (1.680 hours) is 60 EC. |

The other terms have the meanings ascribed to them by the WHW.

## 2. Study programme structure

### Article 2.1 Structure of academic year and units of education

| | | |
|---|---|---|
| 1. | The study programme will be offered in a year divided into two semesters. | Ordinance CvB, see Appendix III |
| 2. | Every semester consists of three consecutive periods. The first two periods each consist of eight weeks, and the final period consists of four weeks. | Ordinance CvB, see Appendix III |
| 3. | A unit of education comprises 6 EC or a multiple thereof. | Ordinance CvB, see Appendix III |
| 4. | By way of exception to paragraph 3, the Executive Board may in special cases and on request of the Faculty Board, stipulate that a unit of education comprises 3 EC or a multiple thereof. | Ordinance CvB, see Appendix III |

## 3. Assessment and examination

### Article 3.1 Signing up for education and examinations

| | |
|---|---|
| 1. Every student must sign up to participate in the units of education of the programme, the examinations and resits. The procedure for signing up is described in an annex to the student statute. | Ordinance CvB, see Appendix III |
| 2. Signing up may only take place in the designated periods. | Ordinance CvB, see Appendix III |

### Article 3.2 Type of examination

| | |
|---|---|
| 1. At the examiner's request, the Examination Board may permit a different form of examination than is stipulated in the study guide. | Advice OLC, approval FGV (7.13 l) |

### Article 3.3 Oral examinations

| | |
|---|---|
| 1. No more than one student will be examined orally at a time, unless specified otherwise in part B for the relevant unit of education. | Advice OLC; approval FGV (7.13 l and n) |
| 2. A second examiner is present when an oral examination is being held, unless the Examination Board has determined otherwise. The oral examination takes place at a time and location to be determined by the examiner within the regular timetable, on campus or online. At the request of the student, and with the approval of the examiners, an audio recording can be made. Should the second examiner not be available unexpectedly, an audio recording of the oral exam can be made. This audio recording is retained by the university for the retention period applicable for examinations. | Advice OLC; approval FGV (7.13 l and n) |

### Article 3.4 Determining and announcing results

| | |
|---|---|
| 1. The examiner determines the result of a written examination within fifteen working days. However, the marking deadline for theses and final assignments is no longer than twenty working days after submission. The examiner will then immediately ensure that the marks are registered and also ensures that the student is immediately notified of the mark, taking due account of the applicable confidentiality standards. | Ordinance CvB, see Appendix III |
| 2. The examiner determines the result (i.e. mark) of an oral examination as soon as possible, but at the latest within one working day after the examination has finished and informs the student accordingly. The third clause of the first paragraph applies. | Advice OLC; approval FGV (7.13 o) |
| 3. In the case of assessments other than oral or written examinations, the Examination Board determines in advance how and by what deadline the student will be informed of the results. | Advice OLC; approval FGV (7.13 o) |

### Article 3.5 Examination opportunities

| | |
|---|---|
| 1. a. Per academic year, two opportunities to take examinations will be offered for each unit of education.<br>b. By way of exception to a., the options for retaking practical exercises, work placements and theses are detailed in the relevant work placement manual, course manual or teaching regulations. | Ordinance CvB, see Appendix III |
| 2. The most recent mark will apply in the event of a resit. A resit is allowed for both passed and failed units of education. | Ordinance CvB, see Appendix III |

| | |
|---|---|
| 3. The resit for an (partial) examination may not take place within ten working days of the announcement of the results for the original (partial) examination. Exceptions are only possible in period 6. | Advice OLC; approval FGV (7.13 j) |
| 4. The Examination Board may allow a student an extra opportunity to sit an examination if that student:<br>a) lacks only those credits to qualify for their degree; and<br>b) has failed the examination during all the previously offered attempts unless participation in an examination was not possible for compelling reasons.<br>The extra opportunity can only be offered if it concerns a written examination, a paper or a take home examination. This provision excludes the practical exercises and the Master's thesis. Requests for an additional examination opportunity must be submitted to the Examination Board no later than 15 July. If necessary, the method of examination may deviate from the provisions in the study guide. | Ordinance CvB, see Appendix III |
| 5. If a unit of education is no longer offered, at least one additional opportunity will subsequently be provided to sit the examination(s) or parts thereof and a transitional arrangement will be included in section B. | Advice OLC, approval FGV (7.13 j) |

## Article 3.6 Marks

| | |
|---|---|
| 1. (Partial) grades are given on a scale from 1 to 10 with no more than one figure after the decimal point. | Ordinance CvB, see Appendix III |
| 2. A final mark between 5 and 6 will be rounded to the nearest whole number: final marks below 5.5, rounded down; final marks of 5.50 or higher, rounded up. All other final marks will be expressed in whole or half marks. | Ordinance CvB, see Appendix III |
| 3. To pass a given course or unit, a final mark of 6 or higher is required. | Ordinance CvB, see Appendix III |
| 4. The Examination Board can allow the use of symbols rather than numbers, for example: s(atisfactory), g(ood), u(n)s(at)is(factory), etc. In case a student does not take part in any examination, the examiner will register the mark 'ns' (c.q. no show). | Ordinance CvB, see Appendix III |

## Article 3.7 Exemption

| | |
|---|---|
| 1. At the written request of the student, the Examination Board may exempt the student from taking one or more examination components, if they:<br>a. have either passed a unit of education at a university or university of applied sciences (HBO) or a research university (WO) that is equivalent in terms of content and level; or<br>b. have demonstrated through their work and/or professional experience that they have sufficient knowledge and skills with regard to the relevant unit of education. | Advice OLC; approval FGV (7.13 r) |
| 2. The Master's thesis is excluded from this exemption possibility. | Advice OLC; approval FGV (9.38 sub b) |
| 3. a. A maximum of 18 EC for a one-year Master's programme and 36 EC for a two-year Master's programme can be accumulated through granted exemptions. The substituting courses are not included.<br>b. This regulation does not apply if a student is (simultaneously) enrolled in multiple programmes, which lead to multiple degrees, where courses may be part of both programmes. In such a case, the Examination Board may deviate from the regulations under a. | Advice OLC; approval FGV (9.38 sub b) |

### Article 3.8 Validity period for results

| | | |
|---|---|---|
| 1. | The validity period of examinations passed and exemption from examinations is unlimited, unless otherwise specified in Section B. | Legal provision |
| 2. | The validity period of a partial examination is limited to the academic year in which it was sat or until the end of the unit of study concerned, as stipulated for the relevant unit of study in Section B. In exceptional circumstances, an examiner may deviate from this; this must be stated on Canvas at the start of the unit of education. | Advice OLC; approval FGV (9.38 sub b) |

### Article 3.9 Right of inspection and post-examination discussion

| | | |
|---|---|---|
| 1. | Within twenty working days after the announcement of the results of a written examination, but at least ten working days before the resit opportunity for that examination, the student can submit a request to review their graded examination. | Advice OLC; approval FGV (7.13 p and q) |
| 2. | Students can view the questions and assignments set in the written examination, and the standards applied when assessing the examination, within the period specified in 3.9.1. | Advice OLC; approval FGV (7.13 q) |
| 3. | The examiner determines whether the inspection takes place collectively or individually. In all cases, the time and place of the inspection is announced in the course manual or learning environment. | Advice OLC; approval FGV (7.13 q) |
| 4. | Students are entitled to feedback on their academic performance. The manner in which this is provided in each unit of education is described in the course manual. | Advice OLC; approval FGV (7.13 p and q) |

## 4. Academic student counselling and study progress

### Article 4.1 Administration of study progress and academic student counselling

| | | |
|---|---|---|
| 1. | The faculty board is responsible for the correct registration of the students' study results. After the assessment of a unit of education has been registered, every student has the right to inspect the result for that component and also has a list of the results achieved at their disposal in VU.nl Dashboard. | Advice OLC; approval FGV (7.13 u) |
| 2. | Enrolled students are eligible for academic student counselling.<br>Academic student counselling is in any case provided by<br>a. The Student Counsellor<br>b. Student psychologists<br>c. Faculty academic advisers | Advice OLC; approval FGV (7.13 u) |

### Article 4.2 Facilities for students with a disability or impairment

| | | |
|---|---|---|
| 1. | Students with a disability or impairment , can submit a request via VU.nl Dashboard to qualify for one or more special facilities with regard to teaching, practical exercises and examinations. These facilities will accommodate the student's individual disability or impairment as much as possible, but may not alter the quality or degree of difficulty of a unit of education or an examination. In all cases, the student must fulfil the exit qualifications for the study programme. | Advice OLC; approval FGV (7.13 m) |
| 2. | The request referred to in the first paragraph must be accompanied by a statement from a doctor or psychologist. If possible, an estimate should be given of the potential impact on the student's study progress. In case of a chronic disability a single (one time) request suffices. | Advice OLC; approval FGV (7.13 m) |

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|---|---|
| 3. Students who have been diagnosed with dyslexia must provide a statement from a BIG-, NIP- or NVO-registered professional who is qualified to conduct a psychological evaluation. | Advice OLC; approval FGV (7.13 m) |
| 4. The faculty board, or the person acting on behalf of the faculty board, decides on teaching facilities and facilities regarding logistics. The Examination Board will decide on requests for facilities with regard to examinations. | Advice OLC; approval FGV (7.13 m) |
| 5. In the event of a positive decision in response to a request as referred to in paragraph 1, the student can make an appointment with the academic adviser to discuss the details of the provisions. | Advice OLC; approval FGV (7.13 m) |
| 6. A request for one or more facilities can be refused if it would place a disproportionate burden on the organisation or the resources of the faculty or university were it upheld. Any such refusal will be substantiated. | Advice OLC; approval FGV (7.13 m) |
| 7. If the disability or impairment justifies an extension of the examination time, the responsible person on behalf of the Examination Board will register in SAP this entitlement to an extension. If a disability or impairment justifies other measures to be taken, the academic adviser can take the necessary measures. The student can consult the study monitor to check which facilities have been granted to them. | Advice OLC; approval FGV (7.13 m) |
| 8. The decision as referred to in paragraph 5 may specify a limited validity of the facilities granted. | Advice OLC; approval FGV (7.13 m) |

## 5. Hardship clause

### Article 5.1 Hardship clause

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|---|---|
| In instances not regulated by the Teaching and Examination Regulations or in the event of demonstrable extreme unreasonableness or unfairness, the faculty board responsible for the study programme will decide, unless the matter concerned is the responsibility of the Examination Board. | Advice OLC; approval FGV (9.38 sub b) |

## Section B1: Programme-specific section

## 6. General programme information and characteristics

### Article 6.1 Study programme information

| | |
|---|---|
| 1. The programme Computer Security, CROHO number 60802, is offered on a full-time basis. | Advice OLC; approval FGV (7.13 u) |

### Article 6.2 Teaching formats used and modes of assessment

| | |
|---|---|
| 1. The programme uses the teaching formats as specified in the study guide. | Advice OLC; approval FGV (7.13 x) |
| 2. The modes of assessment used per unit of education are specified in the study guide. | Advice OLC; approval FGV (7.13 l) |

### Article 6.3 Academic student counselling

| | |
|---|---|
| 1. The programme offers the following counselling in addition to the student counselling mentioned in Section A:<br>- There is an academic advisor for the Master Computer Security. | Advice OLC; approval FGV (7.13 u) |

## 7. Further admission requirements

### Article 7.1 Intake date(s)

| | |
|---|---|
| 1. The Master's programme starts on 1 September. | Advice OLC, Approval FGV (9.38 sub b) |

### Article 7.2 Admission requirements

| | |
|---|---|
| 1. Admission to the Master's programme is possible for an applicant who has obtained a Bachelor's degree at an institution of academic higher education, which demonstrates the following knowledge, understanding and skills:<br>a. knowledge: Programming, Computer Systems<br>b. understanding: Systems thinking, Algorithmic thinking, Critical thinking<br>c. skills: Programming, Software/Hardware practicals, Collaborative projects, Computer Science research, Writing and presenting technical material, Working independently. | Advice OLC, approval FGV (9.38 sub b) |
| 2. Not applicable (no additional requirements for tracks). | Advice OLC, approval FGV (9.38 sub b) |
| 3. Applicants with a Bachelor's degree obtained at an institution outside the Netherlands may be asked for additional methods to prove that they meet the admission requirements. | Advice OLC, approval FGV (9.38 sub b) |
| 4. Not applicable (concerns Dutch language programmes) | Advice OLC, approval FGV (9.38 sub b) |
| 5. Applicants should demonstrate that they have a sufficient level of proficiency in English by meeting at least one of the following standards: | Advice OLC, approval FGV (9.38 sub b) |

| | |
|---|---|
| - IELTS Academic: an overall score of at least 6.5 plus a minimum score of 6.0 for each subcomponent (Listening, Reading, Writing and Speaking); IELTS test has a validity of 2 years.<br>- TOEFL iBT: an overall score of at least 92 plus a minimum score of 12 for the Listening section, 13 for the Reading section, 21 for the Writing section, and 18 for the Speaking section;<br>TOEFL test has a validity of 2 years.<br>- Cambridge Certificate of Proficiency in English (CPE) or Cambridge Certificate of Advanced English (CAE) with an overall score of at least 180. Cambridge tests are valid indefinitely. | |
| 6. Applicants who have completed one of the following meet the English language requirement and therefore are exempted from taking an English language test (as mentioned in the section 5) if:<br>- a secondary education or higher education degree fully taught in English has been completed in Australia, Canada (except Quebec province), Ireland, New Zealand, Singapore, South Africa, the United Kingdom or the United States of America;<br>- a Bachelor's or Master's degree in an English-taught programme accredited by NVAO in the Netherlands has been earned; or<br>- a Bachelor's or Master's degree in an accredited fully English-taught university programme has been earned in a member state of the European Union/ European Economic Area; or<br>- a VWO diploma or equivalent diploma in which the course of English of a comparable level is required has been earned. | Advice OLC, approval FGV (9.38 sub b) |
| 7. The Admission Board will investigate whether the applicant meets the admission requirements. | Advice OLC; approval FGV (9.38 sub b) |

## Article 7.3 Selection criteria

| | |
|---|---|
| 1. In addition to the admission requirements referred to in Article 7.2, the Faculty Board also sets (at least two of) the following selection criteria:<br>a) A high level of relevant knowledge and skills demonstrated by:<br>- Strong academic performance in those parts related to Computer Systems of the Bachelor's degree program, from a high-quality academic institution<br>- A high grade for the Bachelor's thesis or similar, from a high-quality academic institution<br>b) Motivation for the programme, demonstrated by a letter stating the motivation for the programme / field of study / Computer Systems research<br>c) A strong command of research methods from the relevant field, demonstrated by a high grade for research subjects, or comparable, from a high-quality academic institution<br>d) A high general academic level demonstrated by:<br>- A high grade for the Bachelor's thesis or similar from a high-quality academic institution<br>- A scientific article / paper, and open-source software artifacts or FAIR data, not older than 2 years. | Advice OLC, approval FGV (9.38 sub b) |

### Article 7.4 Pre-Master's programme

| | |
|---|---|
| 1. Not applicable | Advice OLC; approval FGV (9.38 sub b) |

## 8. Examinations and results

### Article 8.1 Sequence of examinations

| | |
|---|---|
| 1. Students may participate in the units of education listed below only if they have passed the examination or examinations for following units of education:<br>- XM_0123 Master Project Computer Security after accumulating a minimum of 84 EC from MSc Computer Security courses. | Advice OLC; approval FGV (7.13 h, s & t) |

### Article 8.2 Validity period for results

| | |
|---|---|
| 1. The validity period of the examinations and exemptions from examinations below is limited, as follows: See Article 3.8 of the Teaching and Examination Regulations. No further specific provisions. | Advice OLC; approval FGV (7.13 k) |
| 2. A student may request the Examination Board to extend the validity of an exam. If the student's knowledge is demonstrably outdated, or if the student's skills and understanding are demonstrably outdated, the Examination Board may impose a supplementary examination, impose a replacement examination or refuse to extend the period of validity. | Legal provision |
| 3. In situations where a limited period of validity applies, the period of validity of examinations may be extended in the event of extenuating circumstances as stipulated in WHW Article 7.51, paragraph 2, with at least the period of allocated financial assistance specified in WHW Article 7.51, paragraph 1. | Legal provision |

## Section B2: Programme-specific section – content of programme

### 9. Programme objectives, tracks/specialisations, exit qualifications and language

#### Article 9.1 Workload

| | |
|---|---|
| 1. The programme has a workload of 120 EC. | Advice OLC; (7.13 a) |

#### Article 9.2 Tracks and/or specialisations

| | |
|---|---|
| 1. The programme has the following tracks and/or specialisations: Not applicable | Approval OLC (7.13 b) |

#### Article 9.3 Programme objective

| | |
|---|---|
| 1. The programme aims to:<br>- provide students with the knowledge of the concepts, methods and skills that are relevant to the operation and implementation of secure computer systems, including topics such as programming, computer systems, computer networks, software security, hardware security, network security, and program analysis.<br>- offer students a solid basis for a career in the industry at an academic level;<br>- prepare students for further education to become a scientific researcher.<br>- provide students with an understanding of the position of Computer Security in a broader societal and scientific context. | Advice OLC; (7.13 a) |

#### Article 9.4 Exit qualifications

| | |
|---|---|
| 1. At all events, a graduate of the study programme will:<br>- have solid scientific knowledge and understanding of computer security with a “systems” emphasis, including knowledge of programming, computer systems, software security, hardware security, and program analysis (Knowledge and understanding)<br>- have acquired practical skills in relevant areas of computer security at an academic level (Knowledge and understanding, Applying knowledge and understanding, Making judgements)<br>- be aware of the applications of computer science in general and of computer security in particular, and be capable of applying their knowledge and skills to new and unfamiliar problems (Knowledge and understanding, Applying knowledge and understanding, Making judgements)<br>- be capable of performing effectively in professional situations where scientific knowledge and skills in the field of computer security are required (Communication, Lifelong learning skills)<br>- be capable of communicating at a professional level and of giving clear oral and written presentations of their own work or others’ work (Communication)<br>- be capable of acquiring knowledge, understanding and skills in other sub-areas of computer security within a reasonable period (Lifelong learning skills)<br>- be capable of designing a research plan or project on the basis of a realistic problem description from the field of computer security (Lifelong learning skills)<br>- be capable of conducting scientific research, both independently and as a member of a small team (Communication, Lifelong learning skills)<br>- be capable of consulting international professional literature on relevant areas of computer security and of using the discussions and results found there (Lifelong learning skills)<br>- be capable of formulating, analysing and evaluating scientific results, and of | Approval OLC (7.13 ) |

| | |
|---|---|
| using them to draw relevant conclusions (Lifelong learning skills)<br>- have developed a critical scientific attitude and an awareness of the social and ethical aspects of computer security (Lifelong learning skills)<br>- be thoroughly prepared for an academic programme at PhD level and/or other post-graduate programmes (Lifelong learning skills) | |
| 2. Not applicable | Approval OLC (7.13 b) |
| 3. Language proficiency may be taken into account in the assessment of examinations. | Approval OLC (7.13 c) |

### Article 9.5 Language of instruction

| | |
|---|---|
| 1. The language of instruction is English. | Approval OLC (9.18) |
| 2. The 'Gedragscode vreemde taal' (Code of conduct for foreign languages) applies. | VU policy |

## 10. Curriculum structure

### Article 10.1 Composition of the programme

| | |
|---|---|
| 1. The programme comprises at least a package of compulsory components and an individual Master's thesis or academic work placement. | Ordinance CvB, see Appendix III |
| 2. Additionally the programme can offer:<br>- practical exercises<br>- electives | Advice OLC (7.13 a) |
| 3. Units of education are categorised as specialised (400), research-oriented (500) and highly specialised (600) level. | Ordinance CvB, see Appendix III |

### Article 10.2 Compulsory units of education

*A detailed description per unit of education can be found in the Study Guide.*

| **Unit of education** | **course code** | **nr of EC** | **level** | Advice OLC; (7.13 a) |
|---|---|---|---|---|
| *Year 1* | | | | |
| Programming Large-scale Parallel Systems | XM_40017 | 6 | 400 | |
| Software Security | X_400127 | 6 | 400 | |
| Systems Seminar | XM_0122 | 6 | 600 | |
| Binary and Malware Analysis | X_405100 | 6 | 500 | |
| | | | | |
| *Year 2* | | | | |
| Master Project Computer Security | XM_0123 | 30 | 500 | |
| Advanced Operating Systems | XM_40014 | 6 | 500 | |
| Hardware Security | XM_40019 | 6 | 500 | |

### Article 10.3 Elective units of education

| 1. The student can take one or more of the following electives without prior consent from the Examination Board: | | | | Advice OLC; (7.13 a) |
|---|---|---|---|---|
| **Name of unit of education** | **course code** | **nr of EC** | **level** | |
| *Constrained choice FCC* | | | | |
| Advanced Algorithms | XMU_0060 | 6 | 400 | |

| | | | | |
|---|---|---|---|---|
| Advanced Logic | X_405048 | 6 | 500 | |
| Foundations of Cryptography | XM_0174 | 6 | 400 | |
| Distributed Algorithms | X_400211 | 6 | 500 | |
| Verification for Security | XM_0099 | 6 | 400 | |
| | | | | |
| *Free Electives* | | | | |
| Research Project in Computer Security | XM_0125 | 6 | 500 | |
| Large research Project in Computer Security | XM_0126 | 12 | 600 | |
| Introduction to Computational Science | XMU_418111 | 6 | 400 | |
| Evolutionary Computing | X_400111 | 6 | 400 | |
| Service Oriented Design | X_405061 | 6 | 400 | |
| Green Lab | X_418158 | 6 | 500 | |
| Dynamic Programming and Reinforced Learning | XM_0093 | 6 | 400 | |
| Network Security | XM_0100 | 6 | 400 | |
| Digital Architecture | XM_0127 | 6 | 400 | |
| Distributed Systems | X_400130 | 6 | 400 | |
| The Social Web | X_405086 | 6 | 400 | |
| Performance of Networked Systems | X_405105 | 6 | 400 | |
| High Performance Computing and Big Data | XMU_40013 | 6 | 400 | |
| Security Experiments and Measurements | XM_0098 | 6 | 400 | |
| Advanced Compiler Construction | XM_0139 | 6 | 400 | |
| Secure and Trustworthy Machine Learning | XM_0179 | 6 | 500 | |
| Parallel Programming Practical | X_400162 | 6 | 400 | |
| Cryptographic Engineering | XMU_0047 | 6 | 400 | |
| Web Services and Cloud-based Systems | XMU_418110 | 6 | 400 | |
| Experimental Design and Data Analysis | X_405078 | 6 | 400 | |
| Data Mining Techniques | X_400108 | 6 | 400 | |
| Software Testing | X_400439 | 6 | 400 | |
| AI for Security | XM_0178 | 6 | 400 | |
| Machine Learning for the Quantified Self | XM_40012 | 6 | 500 | |
| 2. If the student wishes to take a different unit of education than listed, advance permission must be obtained in writing from the Examinations Board. | | | | |

### Article 10.4 Practical exercise

Not applicable.

### Article 10.5 Participation in practical training and tutorials

| | |
|---|---|
| 1. In the case of practical exercises, there may be attendance requirements. For details on attendance requirements, please refer to the study guide of the relevant unit of study. | Approval OLC (7.13 d) |
| 2. In the case of tutorials, there may be attendance requirements. For details on attendance requirements, please refer to the study guide of the relevant unit of study. | Approval OLC (7.13 d) |
| 3. For thesis and internship, the programme applies the guidelines as stipulated in the | Approval OLC (7.13 d) |

| | |
|---|---|
| Thesis and Placement Regulations. | |

## 11. Evaluation and transitional provisions

### Article 11.1 Evaluation of the programme

| | |
|---|---|
| 1. The education provided in this programme is evaluated in accordance with the document 'Kwaliteitszorg Onderwijs BETA'. | Approval OLC (7.13 a1) |

### Article 11.2 Transitional provisions

| | |
|---|---|
| 1. By way of departure from the Teaching and Examination Regulations currently in force, the following transitional provisions apply for students who started the programme under a previous set of Teaching and Examination Regulations:<br>- The course Software Threat Analysis (XM_0124) changed its name and focus to AI for Security (XM_0178). Students who successfully completed STA in a previous year are allowed to use this as an elective course, and are also allowed to take the new AI for Security course as another elective.<br>- The course Security and Machine Learning (XM_0135) changed its name to Secure and Trustworthy Machine Learning (XM_0098). Students who successfully completed SML in a previous year are allowed to use this as an elective course, but this would prevent them from taking the new course STML as part of their curriculum.<br>- From academic year 2024-2025, Storage Systems (XM_0092) (formerly a compulsory year 2 course) is no longer offered. Students who started the Master in 2024-2025 (or before) and did not pass Storage Systems (XM_0092) must complete Advanced Operating Systems (XM_40014) instead.<br>- From academic year 2025-2026 Coding and Cryptography (X_405041) is no longer offered. Students who successfully completed CC in a previous year are allowed to use this as an elective course.<br>- Students who successfully completed XB_0003 Compiler Construction cannot take XM_0139 Advanced Compiler Construction.<br>- The elective course Industrial Internship XM_405080 has been removed as per 2024-2025. Students who successfully completed this course in a previous year are allowed to use this as an elective course. | Advice OLC (7.13 a) |

Approval of the Programme Committee of the curriculum structure (submitted via UAS), on 13/12/2024.

Advice and/or approval by the Programme Committee (with the exception of the curriculum structure), on 20/06/2025.

Approved by the Faculty Joint Assembly, on 25/07/2025.

Adopted by the board of the Faculty of Science on 11/08/2025.

## Appendix I

## Overview of articles that must be included in the OER

*Based on Article 7.13, paragraph 2 and Article 7.30b, paragraph 2 of the WHW.*

Section A: Faculty section

| | |
|---|---|
| **2. Study programme structure** | |
| Article 2.1 Structure of academic year and units of education | 7.13 paragraph 2 sub e |
| **3. Assessment and examination** | |
| Article 3.2 Type of examination | 7.13 paragraph 2 sub h, l, j |
| Article 3.3 Oral examinations | 7.13 paragraph 2 sub l, n |
| Article 3.4 Determining and announcing results | 7.13 paragraph 2 sub o |
| Article 3.5 Examination opportunities | 7.13 paragraph 2 sub h, j |
| Article 3.7 Exemption | 7.13 paragraph 2 sub r |
| Article 3.8 Validity period for results | 7.13 paragraph 2 sub k |
| Article 3.9 Right of inspection and post-examination discussion | 7.13 paragraph 2 sub p, q |
| **4. Academic student counselling and study progress** | |
| Article 4.1 Administration of study progress and academic student counselling | 7.13 paragraph 2 sub u |
| Article 4.2 Facilities for students with a disability | 7.13 paragraph 2 sub m |

Section B1: Programme specific – general provisions

| | |
|---|---|
| **6. General programme information and characteristics** | |
| Article 6.1 Study programme information | 7.13 paragraph 2 sub i, r |
| Article 6.2 Teaching formats used and modes of assessment | 7.13 paragraph 2 sub l, x |
| [keuze:] Article 6.3 Academic student counselling | 7.13 paragraph 2 sub u |
| **7. Further admission requirements** | |
| Article 7.2 Admission requirements | 7.30b paragraph 1 |
| Article 7.3 Selection criteria | 7.30b paragraphs 2 & 5 |
| Article 7.4 Pre-Master's programme | 7.57i section 1 |
| **8. Examinations and results** | |
| Article 8.1 Sequence of examinations | 7.13 paragraph 2 sub h, s, t |
| [option 1:] Article 8.2 Validity period for results | 7.13 paragraph 2 sub k |
| [option 2:] Article 8.2 Validity period for results | 7.13 paragraph 2 sub k |

Section B2: Programme-specific section – content of programme

| | |
|---|---|
| **9. Programme objectives, tracks/specialisations, exit qualifications and language** | |
| Article 9.1 Workload | 7.13 paragraph 2 sub g |
| Article 9.2 Tracks and/or specialisations | 7.13 paragraph 2 sub a |
| Article 9.3 Programme objective | 7.13 paragraph 2 sub a |
| Article 9.4 Exit qualifications | 7.13 paragraph 2 sub b, c |
| Article 9.5 Language of instruction | 9.18 (*implementation expected in 2020*) |
| **10. Curriculum structure** | |
| Article 10.1 Composition of the programme | 7.13 paragraph 2 sub a |
| Article 10.2 Compulsory units of education | 7.13 paragraph 2 sub a |
| [Keuze:] Article 10.3 Elective units of education | 7.13 paragraph 2 sub a |
| [Keuze:] Article 10.4 Practical exercise | 7.13 paragraph 2 sub d |
| Article 10.5 Participation in practical training and tutorials | 7.13 paragraph 2 sub d |
| **11. Evaluation and transitional provisions** | |
| Article 11.1 Evaluation of the programme | 7.13 paragraph 2 sub a1 |
| Article 11.2 Transitional provisions | 7.13 paragraph 2 sub a |

## Appendix II

## Overview of advisory and approval rights of Programme Committees (OLC) and Faculty Joint Assembly (FGV)

| Article 7.13, paragraph 2, of the Higher Education and Research Act | FGV | | OplC | |
|---|---|---|---|---|
| | I | A | I | A |
| a. the content of the programme and the associated examinations | | | | ■ |
| a1. the manner in which teaching and education in the relevant programme are evaluated | | | ■ | |
| b. the content of the specialisations offered as part of the programme | | | ■ | |
| c. the programme's final attainment levels with regard to the knowledge, understanding and skills | | | ■ | |
| d. where applicable, the design of practical exercises | | | ■ | |
| e. the workload of the programme and of each of its constituent units of education | | | ■ | |
| f. the detailed rules referred to in Article 7.8b, sixth paragraph, and Article 7.9, fifth paragraph (recommendation on continuation of studies) | | | | ■ |
| g. the Master's programmes to which Article 7.4a, eighth paragraph, applies (elevated workload) | | | ■ | |
| h. the number and sequence of examinations and the times at which these can be taken | ■ | | | ■ |
| i. the full-time, part-time or work-study structure of the programme | ■ | | | ■ |
| j. where necessary, the order in which students can take the examinations and the final examinations for Bachelor's and Master's programmes, as well as the timing of and number of opportunities | ■ | | | ■ |
| k. where necessary, the period of validity for pass grades awarded for examinations, notwithstanding the authority of the Examination Board to extend this period of validity | ■ | | | ■ |
| l. whether examinations are administered in oral, written or another form, notwithstanding the authority of the Examination Board to decide otherwise | ■ | | | ■ |
| m. the way in which students with a disability or chronic health condition are given a reasonable opportunity to take the examinations | ■ | | | ■ |
| n. the public nature of oral examinations, subject to the right of the Examination Board to determine otherwise in special cases | ■ | | | ■ |
| o. the period within which the results of an examination must be announced, together with details of whether this period can be altered and if so in what way | ■ | | | ■ |
| p. the way in which and the period within which students who have taken an examination are given the opportunity to inspect their marked work | ■ | | | ■ |
| q. the way in which and the period within which information can be provided about the questions asked and exercises given in the framework of a written examination and about the standards used for assessment | ■ | | | ■ |
| r. the grounds on which the Board of Examiners may grant exemptions from taking one or more examinations on the basis of examinations and final examinations for Bachelor's and Master's programmes previously completed in higher education, or on the basis of knowledge or skills acquired outside higher education | ■ | | | ■ |
| s. where necessary, the stipulation that students must pass certain examinations as a condition for admission to other examinations | ■ | | | ■ |
| t. where necessary, the obligation to take part in certain practical exercises with a view to admission to the examination in question, subject to the authority of the Examination Board to grant exemption from this obligation, with or without the imposition of alternative requirements | ■ | | | ■ |
| u. the monitoring of academic progress and individual student support and guidance | ■ | | | ■ |
| v. where applicable, the manner in which students are selected for a special track within a programme as referred to in Article 7.9b (excellence track within a programme) | | | ■ | |
| x. the actual design of the education provided | ■ | | | ■ |
| ***All other matters that are regulated in the Academic and Examination Regulations but which are not mentioned as such in Article 7.13 of the Higher Education and Research Act under points a to x.*** | | | | ■ |

*The lettering corresponds to the lettering of Article 7.13 section 2 of the WHW*

**Abbreviations used:**

FGV: Faculty Joint Assembly
OplC: Programme Committee
I: right of approval
A: right of advice

## Appendix III
## Ordinances of VU Executive Board (CvB) and Binding Master's Guidelines

| Section A, article: | Concerns: | CvB ordinance / guideline |
|---|---|---|
| 2.1.1, 2.1.2 | Year planning two semesters 8-8-4 (uniform year calendar VU-UvA) | 29-9-2008 (period 2009-2015)<br>22-05-2014 (period 2016-2025) |
| 2.1.3, 2.1.4 | Units of education | Bachelor's and Master's education guidelines included in model TER, revised on 2 July 2019 |
| 3.1 | Signing up for education and examinations | CvB ordinance 17-11-2020, prior consent of USC. |
| 3.4.1 | Determination and publication of the results<br>(1) Grading deadline exams ten working days<br>(2) Theses twenty working days | (1) Bachelor's guideline included in model TER, revised on 2 July 2019<br>((2) Quality demand 11 from the VU-wide assessment policy, CvB ordinance 15-05-2012, approval GV 28-02-2013 |
| 3.5.1 | Two possibilities to take examinations per year | Bachelor's and Master's education guidelines included in model TER, revised on 2 July 2019 |
| 3.5.2 | Resit: most recent grade is valid, resit allowed in case of pass grade | Taken from the UvA guidelines, as part of the harmonisation, CvB ordinance 24-02-2014 |
| 3.5.4 | One-time extra resit in relation to examination requirement | Included in (prior) model OER 16-17 following a request from committee O&O and adopted by CvB op 27-10-2015 |
| 3.6 | Grades | CvB ordinance 30-09-2010, prior consent USR. As a result of harmonisation UvA, the guideline: 5.5 is a pass, has been added. CvB ordinance 24-02-2014. |
| **Section B1, article:** | **Concerns:** | **CvB ordinance / guideline** |
| 7.2 | Admission criteria | Bachelor's and Master's education guidelines included in model TER, revised on 2 July 2019 |
| 7.3 | Selection criteria | Bachelor's and Master's education guidelines included in model TER, revised on 2 July 2019 |
| **Section B2, article:** | **Concerns:** | **CvB ordinance / guideline** |
| 10.1.1 | Composition programme | Bachelor's and Master's education guidelines included in model TER, revised on 2 July 2019 |
| 10.1.3 | Categorisation of units | Bachelor's and Master's education guidelines included in model TER, revised on 2 July 2019 |

## Appendix IV
## Higher Education and Research (Implementation) Act

1.The extenuating personal circumstances referred to in Article 7.8b, paragraph 3 and 7.9, paragraph 3 of the Act (WHW) are limited to:

a. illness of the person concerned,
b. physical, sensory or other impairment of the person concerned,
c. pregnancy of the person concerned,
d. extenuating family circumstances,
e. membership, including the chairmanship of:

1. universities: the university council, faculty council, the body established under the participation regulation referred to in Article 9.30, paragraph 3, or Article 9.51, paragraph 2, of the Act, the programme management or the programme committee, or membership on the board of a foundation whose bylaws allow for the exploitation of facilities belonging to the student services, or an equivalent body with regard to its activities in the opinion of the board of the institution,

2. universities of applied sciences: the participation council, district council, student committee or programme committee,

f. other circumstances to be designated by the board of the institution in the regulation as referred to in Article 7.8b, paragraph 6, and Article 7.9, paragraph 5, of the Act in which the person concerned engages in activities within the framework of the organisation and the administration of the affairs of the institution,
g. membership on the board of a student organisation of a certain size with full legal capacity, or a similar organisation of a certain size, whose primary task regards general societal interest and which actually develops activities for this purpose,
h. other personal circumstances set out in the Teaching and Examination Regulations as referred to in Article 7.13 of the Act, pursuant to Article 7.13, paragraph 2, clause f of the Act,
i. personal circumstances other than those referred to in a – h above which, if overlooked by the governing bodies of the institution, would lead to an obviously unfair outcome.

2 The institutional board may, for the purposes of the first paragraph, part g, establish specific rules regarding the maximum number of eligible board members per organisation per academic year, as well as the eligible administrative offices.